

# Numerical Analysis homework II

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Due time: October 23, 2025

## Abstract

The abstract is not necessary for the theoretical homework, but for the programming project, you are encouraged to write one.

## I. Determine the remainder

By Newton interpolation  $\forall x \in (x_0, x_1)$ :

$$f(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x](x - x_0)(x - x_1)$$

So, we get:

$$\frac{f''(\xi(x))}{2} = f[x_0, x_1, x]$$

i.e.

$$\frac{1}{(\xi(x))^3} = \frac{f[x_0, x_1] - f[x_1, x]}{x_0 - x} = \frac{\frac{f[x_0] - f[x_1]}{x_0 - x_1} - \frac{f[x_1] - f[x]}{x_1 - x}}{x_0 - x} = \frac{\frac{1}{2x} - \frac{1}{2}}{1 - x} = \frac{1}{2x}$$

so  $\xi(x) = (2x)^{\frac{1}{3}}$ .

Easily to continuously extend an easy function like  $\xi(x)$  to the closure to its domain.

So  $\xi(x)_{\max} = 4^{\frac{1}{3}}, \xi(x)_{\min} = 2^{\frac{1}{3}}, f''(\xi(x))_{\max} = \frac{1}{2x_{\max}} = \frac{1}{2}$ .

## II. Interpolation of Polynomials in $\mathbb{P}_{2n}^+$

May as well  $x_0 < x_2 < \dots < x_n$ , we choose

$$\pi_k(x) = \prod_{i=0, i \neq k}^n \frac{(x - x_i)^2}{(x_k - x_i)^2} \in \mathbb{P}_{2n}^+$$

obviously:

$$\begin{cases} \pi_k(x_i) = 1, i = k \\ \pi_k(x_i) = 0, i \neq k \end{cases}$$

then:

$$p(x) = \sum_{k=0}^n f_k \pi_k(x) \in \mathbb{P}_{2n}^+, (0 \leq f_k)$$

and  $\forall k \in \{0, 1, \dots, n\}, p(x_k) = f_k \pi_k(x_k) = f_k$ .

## III. The remainder of $e^x$

### III.a

When  $n = 1$ , LHS  $= f[t, t+1] = \frac{e^{t+1} - e^t}{t+1-t} = (e-1)e^t = \text{RHS}$ . We suppose  $n = k$  has done, then:

$$f[t, t+1, \dots, t+k+1] = \frac{f[t+1, t+2, \dots, t+k+1] - f[t, t+1, \dots, t+k]}{t+k+1-t} = \frac{(e-1)^k (e^{t+1} - e^t)}{(k+1)k!} = \text{RHS}$$

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### III.b

By III.a we know

$$\frac{(e-1)^n e^0}{n!} = \frac{1}{n!} f^{(n)}(\xi) = \frac{1}{n!} e^\xi \Rightarrow \xi = n \ln(e-1) > 0.54n$$

So  $\xi$  located to the right of the midpoint  $n/2$ .

## IV. Interpolation of given points

### IV.a

By Newton Interpolation, we get  $f[0] = 5, f[1] = 3, f[3] = 5, f[4] = 12$ .

$$f[0, 1] = -2, f[1, 3] = 1, f[3, 4] = 7.$$

$$f[0, 1, 3] = 1, f[1, 3, 4] = 2$$

$$f[0, 1, 3, 4] = \frac{1}{4}. \text{ So } p_3(f; x) = 5 - 2x + x(x-1) + \frac{1}{4}x(x-1)(x-3).$$

### IV.b

By IV.a, we know

$$p'_3(f; x) = -3 + 2x + \frac{1}{4}[(x-1)(x-3) + x(x-3) + x(x-1)] = -\frac{9}{4} + \frac{3}{4}x^2$$

when  $p'_3(f; x) = 0, \Rightarrow x = \sqrt{3}, -\sqrt{3}$ , and  $p''_3(f; x) = \frac{3}{2}x > 0, (x \in (1, 3))$ . So, we get  $\sqrt{3}$  is a approximate value of  $x_{\min}$

## V. Interpolation of given points

### V.a

$$f[0] = 0, f[1] = 1, f[2] = 2^7$$

$$f[0, 1] = 1, f[1, 1] = 7, f[1, 2] = 2^7 - 1, f[2, 2] = 7 * 2^6$$

$$f[0, 1, 1] = 6, f[1, 1, 1] = 21, f[1, 1, 2] = 2^7 - 8, f[1, 2, 2] = 5 * 2^6 + 1$$

$$f[0, 1, 1, 1] = 15, f[1, 1, 1, 2] = 2^7 - 29, f[1, 1, 2, 2] = 3 * 2^6 + 9$$

$$f[0, 1, 1, 1, 2] = 2^6 - 22, f[1, 1, 1, 2, 2] = 2^6 + 38$$

$$f[0, 1, 1, 1, 2, 2] = 30$$

### V.b

$$f^{(5)}(\xi) = \frac{7!}{2*5!}\xi^2 = 30, \text{ so } \xi^2 = \frac{10}{7}, \text{ so } \xi = \sqrt{\frac{10}{7}}$$

## VI. Hermite Interpolation of given points

### VI.a

$$f[0] = 1, f[1] = 2, f[3] = 0$$

$$f[0, 1] = 1, f[1, 1] = -1, f[1, 3] = -1, f[3, 3] = 0$$

$$f[0, 1, 1] = -2, f[1, 1, 3] = 0, f[1, 3, 3] = \frac{1}{2}$$

$$f[0, 1, 1, 3] = \frac{2}{3}, f[1, 1, 3, 3] = \frac{1}{4}$$

$$f[0, 1, 1, 3, 3] = -\frac{5}{36}$$

$$\text{So, } f(2) = 1 + (2-0) - 2(2-0)(2-1) + \frac{2}{3}(2-0)(2-1)^2 - \frac{5}{36}(2-0)(2-1)^2(2-3) = 1 + 2 - 4 + \frac{4}{3} + \frac{5}{18} = \frac{11}{18}$$

### VI.b

By Hermite Interpolation, we know

$$f(x) - p(f; x) = f[0, 1, 1, 3, 3, x]x(x-1)^2(x-3)^2 = \frac{f^{(5)}(\xi(x))}{5!}x(x-1)^2(x-3)^2$$

Considering  $x(x-1)^2(x-3)^2_{\max} \approx 2.0594$  So the error

$$|f(x) - p(f; x)| \leq 2.06 \frac{M}{5!}$$

## VII. Prove equation

We prove by induction, when  $k = 1$ ,  $\Delta f(x) = hf[x_0, x_1]$ ,  $\nabla f(x) = hf[x_0, x_{-1}]$ . Assume that  $k = n$  has been proved, when  $k = n + 1$ :

$$\Delta^{k+1}f(x) = \Delta^k f(x+h) - \Delta^k f(x) = k!h^k f[x_1, x_2, \dots, x_{k+1}] - k!h^k f[x_0, x_2, \dots, x_k]$$

then By Newton Interpolation:

$$\Delta^{k+1}f(x) = (k+1)!h^{k+1}f[x_0, x_1, \dots, x_{k+1}]$$

the same way for  $\nabla$  Assume that  $k = n$  has been proved, when  $k = n + 1$ :

$$\nabla^{k+1}f(x) = \nabla^k f(x) - \nabla^k f(x-h) = k!h^k f[x_0, x_{-1}, \dots, x_{-k}] - k!h^k f[x_{-1}, x_{-2}, \dots, x_{-k-1}]$$

then By Newton Interpolation:

$$\nabla^{k+1}f(x) = (k+1)!h^{k+1}f[x_0, x_{-1}, \dots, x_{-k-1}]$$

## VIII. Prove equation

By defination of  $\frac{\partial f}{\partial x_0}$ , we know

$$\frac{\partial f[x_0, x_1, \dots, x_n]}{\partial x_0} = \lim_{\Delta x \rightarrow 0} \frac{f[x_0 + \Delta x, x_1, \dots, x_n] - f[x_0, x_1, \dots, x_n]}{\Delta x} = \lim_{\Delta x \rightarrow 0} f[x_0 + \Delta x, x_0, x_1, \dots, x_n]$$

And  $f$  is differentiable at  $x_0$ , so  $f$  is continuous at  $x_0$ , so with finite steps, the error simulate in progress will control by a const  $c$  and  $\varepsilon$  by the continuous at  $x_0$ .i.e.

$$|f[x_0 + \Delta x] - f[x_0]| < \varepsilon, |f[x_0 + \Delta x, x_0] - f[x_0, x_0]| < \varepsilon$$

we can prove by induction, every interpolation has only a  $c\varepsilon$  difference, in which  $c$  is a constant  $\Rightarrow$

$$\lim_{\Delta x \rightarrow 0} f[x_0 + \Delta x, x_0, x_1, \dots, x_n] = f[x_0, x_0, x_1, \dots, x_n]$$

## IX. Min-Max problem

i.e. to solve:

$$\min_{a_i \in \mathbb{R}, i \in 1, 2, \dots, n} \max_{x \in [a, b]} |a_0 x^n + a_1 x^{n-1} + \dots + a_n| = |a_0| \min_{a_i \in \mathbb{R}, i \in 1, 2, \dots, n} \max_{x \in [a, b]} |x^n + \frac{a_1}{a_0} x^{n-1} + \dots + \frac{a_n}{a_0}|$$

$\Rightarrow$

$$|a_0| \min_{a_i \in \mathbb{R}, i \in 1, 2, \dots, n} \max_{x \in [a, b]} |x^n + a_1 x^{n-1} + \dots + a_n| = |a_0| \min_{a_i \in \mathbb{R}, i \in 1, 2, \dots, n} \max_{x \in [-\frac{b-a}{2}, \frac{b-a}{2}]} |x^n + a_1 x^{n-1} + \dots + a_n|$$

Notice that  $a < b$  otherwise the value is trivial  $\Rightarrow$

$$\begin{aligned} |a_0| \min_{a_i \in \mathbb{R}, i \in 1, 2, \dots, n} \max_{x \in [-1, 1]} |(\frac{b-a}{2})^n x^n + (\frac{b-a}{2})^{n-1} a_1 x^{n-1} + \dots + a_n| &= (\frac{b-a}{2})^n |a_0| \min_{a_i \in \mathbb{R}, i \in 1, 2, \dots, n} \max_{x \in [-1, 1]} |x^n + a_1 x^{n-1} + \dots + a_n| \\ &\geq (\frac{b-a}{2})^n |a_0| \min_{a_i \in \mathbb{R}, i \in 1, 2, \dots, n} \frac{1}{2^{n-1}} = |a_0| \frac{(b-a)^n}{2^{2n-1}} \end{aligned}$$

In which the last " $\geq$ " is by the corollary in the book. And we know that choose Chebyshev Polynomials can get the equal.

## X. The minimal element in infity norm

If NOT, we assume  $\exists p \in \mathbb{P}_n^a$ . s.t.

$$\|p\|_\infty < \|\hat{p}_n\|_\infty$$

Let  $Q(x) = \hat{p}_n(x) - p(x)$ , then in the  $n+1$  extreme points ( $\in [-1, 1]$ ) of  $T_n(x)$ , like the proof in Chebyshev Theorem  $Q(x)$  will change the signals  $n+1$  times, so  $Q(x)$  have  $n$  zeros in  $[-1, 1]$ . Meanwhile,  $Q(a) = 1 - 1 = 0$ . So  $Q(x)$  have  $n+1$  zeros, and  $Q(x) \in \mathbb{P}_n$ , so  $Q(x) \equiv 0$ . Contradiction!

## XI. Lemma Proof

$$RHS = \binom{n}{k} t^k (1-t)^{n-k} \frac{n-k}{n} + \binom{n}{k+1} t^{k+1} (1-t)^{n-k-1} \frac{k+1}{n}$$

$$LHS = \binom{n-1}{k} t^k (1-t)^{n-1-k}$$

Notice that  $\binom{n}{k} \frac{n-k}{n} = \frac{n!}{k!(n-k)!} \frac{n-k}{n} = \frac{n!}{(k+1)!(n-k-1)!} \frac{k+1}{n} = \binom{n}{k+1} \frac{k+1}{n} = \frac{(n-1)!}{k!(n-k-1)!} = \binom{n-1}{k}$ . So

$$RHS = \binom{n-1}{k} t^k (1-t)^{n-k-1} (1-t+t) = \binom{n-1}{k} t^k (1-t)^{n-1-k} = LHS$$

## XII. Lemma Proof

$$\int_0^1 b_{n,k}(t) dt = \int_0^1 \binom{n}{k} t^k (1-t)^{n-k} dt = \binom{n}{k} B(k+1, n+1-k) = \binom{n}{k} \frac{\Gamma(k+1)\Gamma(n+1-k)}{\Gamma(n+2)} = \frac{n!k!(n-k)!}{(n-k)!k!(n+1)!} = \frac{1}{n+1}$$